

CONTESTED

A sustainability framework for the S&P 500 that publishes its own uncertainty

Sector note — Initiating framework coverage

S&P 500 Energy · 18 companies · FY2021–FY2025

ETHack 2026 · Challenge #1 · Citadel & Citadel Securities

13 September 2026

We do not publish an ESG score. We publish the distribution of every defensible ESG score, and the width of that distribution is the most useful number nobody reports.

Universe	18 S&P 500 Energy constituents, five Bloomberg BECS peer groups
Inputs	139 fields across four pillars — Environment, Social, Governance, Financial resilience
Method	Specification curve analysis (Simonsohn, Simmons & Nelson, 2020) applied to ESG scoring
Grid	9 discretionary axes → 17,496 defensible specifications per company, per weighting
Headline	Median company moves 7.0 places of 18 across its interquartile range
Settled	2 of 153 pairwise comparisons survive every method. A published league table asserts all 153.
Deliverable	Consensus rank · contestation · the axis that decides it · the disclosure that would settle it

Executive summary

1. The product the market buys is a point estimate of a quantity that has no point estimate. Berg, Kölbel & Rigobon (2022) measured an average correlation of 0.61 between the major ESG raters, against 0.99 for credit ratings. Every provider computes a distribution over method choices, publishes one draw from it, and discards the rest. The discarded part is the information an allocator needs.

2. We did not invent our uncertainty axes. We read MSCI's and Bloomberg's published methodologies end to end and identified thirteen points at which a rater must make a discretionary call. **They differ at seven and agree at six.** We vary the seven where they differ, fix the six where they agree, and declare each fixed assumption. Two further axes are forced by the data itself. The result is a 17,496-cell grid in which every cell is a method one of the two largest providers in the market has published, or a documented variant of one.

3. The disagreement is large and it is not an artefact. The median company moves 7.0 rank places of 18 across its interquartile range. Strip out every branch a critic might call extreme — hold the question to *level*, raise the evidence bar to *majority*, drop the worst-case missing rule, use sector peers — and the median is still **3.6 places**. The spread survives the removal of its own tail.

4. Two negative results, published deliberately. The aggregation function — MSCI's arithmetic mean against Bloomberg's shifted power mean — explains **0.3%** of rank variance. The governance-polarity question explains **0.5%**. We implemented both specifically so that we could report they do not matter here. What does matter is *direction* (18.2%) and *peer set* (13.1%): what question you ask, and who you ask it against. Both are scope choices. Neither is a weight.

5. The output is not a score. Per company we publish four objects: what is **settled** (comparisons that hold under every method), what is **contested**, which single choice is the **pivot**, and which missing disclosure is the **remedy**. That is a strict superset of a rating: an allocator who wants one number can take the consensus and ignore the rest.

EXHIBIT 1 The ranking, and how much of it is a choice

Full grid, 17,496 specifications, equal pillar weights. Consensus is the median rank percentile; contestation is the interquartile width, reported in **rank places out of 18**.

#	Ticker	Peer group	Consensus	Contestation (places)	Verdict	Decided by	Coverage
1	BKR	Services	82	6.0	opinion	peer set (22%)	78%
2	EXE	E&P	82	6.0	opinion	direction (34%)	91%
3	CTRA	E&P	76	7.0	opinion	window (20%)	81%
4	VLO	Refining	71	5.0	opinion	missing data (11%)	77%
5	EOG	E&P	71	10.0	opinion	direction (60%)	90%
6	SLB	Services	65	7.0	opinion	direction (26%)	78%
7	COP	E&P	53	6.0	opinion	direction (20%)	93%
8	OXY	E&P	47	10.0	opinion	direction (38%)	92%
9	XOM	Integrated	47	6.0	opinion	direction (29%)	99%
10	MPC	Refining	41	9.0	opinion	direction (49%)	79%
11	PSX	Refining	41	8.0	opinion	direction (9%)	77%
12	TRGP	Midstream	41	9.0	opinion	peer set (26%)	74%
13	WMB	Midstream	41	11.0	opinion	peer set (13%)	75%
14	KMI	Midstream	41	7.0	opinion	peer set (11%)	76%
15	DVN	E&P	35	6.0	opinion	peer set (32%)	89%
16	CVX	Integrated	35	6.0	opinion	missing data (8%)	97%
17	HAL	Services	18	7.0	opinion	direction (20%)	78%
18	FANG	E&P	18	4.0	opinion	peer set (20%)	91%

Verdict thresholds: robust < 1.5 places, contested < 3.0, opinion ≥ 3.0. Source: ETHack/model/out/summary.csv, pivot.csv.

Read the table as a warning, not a ranking. Under the full grid every company is an opinion. That is the honest result for this sector on this data, and the value is carried by the three columns to the right of the consensus, not by the consensus itself.

1. The problem

1.1 A number that is a choice

Sustainability ratings are consumed as measurements. They are constructed as opinions. The gap is measurable and it is large:

EXHIBIT 2 Inter-rater agreement

Domain	Average pairwise correlation	Source
Credit ratings	0.99	Berg, Kölbel & Rigobon (2022)
ESG ratings, six major providers	0.61	Berg, Kölbel & Rigobon (2022)
ESG ratings, alternative estimate	0.45	Dimson, Marsh & Staunton (2020)

Berg et al. decompose that divergence into three causes: **measurement** (the same concept measured differently, 50.1%), **scope** (what counts at all, 36.7%) and **weights** (13.2%). The finding that governs this paper's design is the last one: **weights are the smallest of the three causes**, and a framework that varies only weights attacks 13% of the problem while leaving 87% untouched.

1.2 Why this matters to an allocator

An allocator does not act on a score; they act on a decision rule — include or exclude, overweight or underweight, engage or divest. A rating is useful exactly insofar as it changes that decision reliably. A rating that moves seven places out of eighteen depending on conventions the provider did not disclose does not change the decision reliably; it changes it arbitrarily, and the arbitrariness is invisible.

The framework in this paper makes it visible, and separates the companies about which every defensible method agrees from the companies about which the rating is an opinion. The first is investable information. The second is a warning label.

2. What we mean by sustainability

This is the largest assumption in the framework and we state it first.

MSCI rates a company's resilience to *financially relevant* ESG risk and frames externalities by whether they become unanticipated costs **for the company** (MSCI, p.3). Bloomberg defines financial materiality as impact on revenue, operating costs, cost of capital, asset value and liabilities (Bloomberg, p.8). Both measure **single materiality**, outside-in: the world's effect on the firm. Neither measures the firm's effect on the world.

Our indicator set does not do this. Scope 1 emissions, embedded carbon in reserves, gas flaring, freshwater withdrawal, spill volume and recordable incident rates are **externality** measures, inside-out. Our financial pillar — leverage, coverage, refinancing wall, reserve life — is closer to the providers' frame, and is included on the grounds that a company which cannot survive its capital structure cannot deliver on a 2050 commitment.

So the framework is a deliberate hybrid, and the challenge brief does not resolve which of the two it wants. That ambiguity is itself a finding: **the single question that determines every downstream choice is not settled by the mandate, and the two incumbents resolved it the same way without being asked to.**

We fix this rather than varying it, for a structural reason. Switching the direction of materiality changes the *indicator set*, not the arithmetic over a fixed indicator set. A specification curve varies the method holding inputs constant; two different indicator sets are two different studies.

3. Data

3.1 Sources

EXHIBIT 3 Data sources

Source	Provides	Licensed
Bloomberg materiality maps, 18 companies, FY2021–FY2025	135 requested fields plus 57 further delivered, across five BECS peer groups, ES and G sheets	Yes — withheld
Own FY2025 credit workbook	leverage, interest coverage, FCF/debt, capex/D&A, payout/FCF, maturities <24m, ROCE, reserve life, three agencies' adjusted leverage	No — SEC XBRL, published ratings
SEC XBRL frames · EPA GHGRP · EPA EEIO v1.3	secondary pipeline, 30 companies across 10 GICS sectors (see §7.2)	No — free, government

Bloomberg Terminal cell values are not redistributed. Everything derived from them — the loader, the formula, every computed output and the working explorer — is published. See `bruh/mat_maps/WITHHELD.md`.

3.2 Universe

Eighteen of the twenty S&P 500 Energy constituents. **EQT and ONEOK have no Bloomberg export and are absent**; we did not pad the sample to a round number.

BECS peer group	n	Members
Exploration & Production	7	COP, CTRA, DVN, EOG, EXE, FANG, OXY
Integrated Oils	2	CVX, XOM
Midstream	3	KMI, TRGP, WMB
Refining & Marketing	3	MPC, PSX, VLO
Oilfield Services & Equipment	3	BKR, HAL, SLB

3.3 The disclosure constraint, which shapes everything downstream

EXHIBIT 4 Field coverage, by pillar and fiscal year (share of cells filled)

Pillar	Fields	FY2021	FY2022	FY2023	FY2024	FY2025
Environment	37	59%	59%	60%	61%	36%
Social	14	65%	65%	65%	66%	39%
Governance	79	97%	97%	97%	97%	92%
Financial	9	94%	94%	94%	94%	94%

Two facts follow, and both became design decisions.

First, FY2025 is barely reported for E and S. Sustainability reporting runs a year or more behind. Reading "the latest column" silently compares a company that has published against one that has not. Bloomberg splits exactly this into two products — *Latest Scores* and *Fiscal Year Scores* (Bloomberg, p.34) — and so do we: the window axis has three options, not two, and one of them holds every company to the last complete year.

Second, the field set is not shared across peer groups.

EXHIBIT 5 Fields by breadth of reporting

Reported by	Total	E	S	G	F
all 18 companies	75	3	1	63	8
12–17 companies	40	18	8	14	0
9–11 companies	12	7	3	1	1
1–8 companies	11	9	2	0	0

Bloomberg builds the materiality map per peer group, so the five sub-industries do not answer the same questions. **Requiring that every company in the sector has answered the same question leaves three environmental fields and one social field.** An eighteen-company comparison of environmental *performance* does not exist in this export. That is not a gap in our pipeline; it is what the sector's disclosure looks like when you demand comparability.

3.4 What is not in the data

There is no revenue and no headcount anywhere in the export. EBITDA is the economic denominator available to us. Bloomberg's own methodology calls revenue a *fallback* for when the physical activity metric is unavailable (Bloomberg, p.17); we are one rung further down that ladder and say so. Production in MMboe/a exists for the nine producers only — so **the correct denominator is unavailable for half the sector**, which is Bloomberg's documented fork made literal.

The credit workbook is a single fiscal year. Under *change* the financial pillar therefore has no trajectory and drops out, and the composite renormalises over the pillars that remain. Holding one year flat and reporting a change of zero would be a fabricated time series.

4. Method

4.1 Where the providers actually choose

We read both methodologies and located every point at which a rater must exercise discretion. **Seven differ; six agree.** We vary the seven and fix the six.

EXHIBIT 6 Provider decision-point map

#	Decision	MSCI	Bloomberg	Same?	Our treatment
1	What is measured	financially material risk to the firm	financially material risk to the firm	agree	fix — and declare our hybrid (\$2)
2	Peer group	GICS Sub-Industry, 8-digit	BECS , a bespoke ESG classification	differ	vary
3	Size denominator	modelled 0–10 exposure, ~80 segment metrics	explicit activity metric; revenue as fallback	differ	vary
4	Raw value → grade	percentile benchmarks	rejects percentiles in print ; log-log residual	differ	vary
5	Missing data	absence scores 0 on <i>management</i>	no imputation; drop, then cap by disclosure	differ	vary
6	Aggregation	weighted arithmetic mean	weighted power mean , $p=\frac{1}{2}$, $s=1$	differ	vary

#	Decision	MSCI	Bloomberg	Same?	Our treatment
7	Pillar weights	impact × horizon; G floored at 33%	analyst rank 1–5; G fixed at rank 3	differ in mechanism	fix at presets + user slider
8	Relative or absolute	E,S relative; G absolute	mostly relative; fatalities absolute	both break their own rule	fix relative, pair with absolutes
9	Time window	3-year controversy lookback	parameters over 3 most recent years	agree on 3 years	vary window and direction
10	Final discretisation	0–10 into 7 letter grades	0–10 into 7 bands	agree; neither justifies 7	fix — we do not discretise
11	Human override	committee overrides and truncations	none; judgement is upstream	differ	fix — none
12	Controversies	central: severity matrix, 3-year lookback	absent entirely	differ	fix — absent, declared
13	Non-public data	refused outright	public only, limited regional estimation	near-agree	fix — public only

Decision 11 is worth isolating. MSCI factors committee-level overrides into the final industry-adjusted score and requires committee approval for exceptions, truncations and two-notch rating changes. **A specification curve requires the method to be a function of the data. You cannot enumerate a committee.** Our model is Bloomberg-shaped by necessity as much as by preference.

4.2 The nine axes

EXHIBIT 7 The specification grid

Axis	Options	n	Provenance
Exposure base	EBITDA · equity · production	3	Bloomberg activity metrics, p.17
Window	latest · fiscal year · 3-year mean	3	Bloomberg Latest vs Fiscal Year, p.34
Direction	level · change	2	Bloomberg Rate-Anchored model, p.18
Polarity	orthodox · neutral · stability	3	15 fields where the two governance conventions disagree
Normaliser	percentile · winsorised · log min–max · elasticity	4	MSCI p.12; Bloomberg pp.41, 45–46
Peer set	BECS group · producers/downstream · all 18	3	MSCI GICS vs Bloomberg BECS
Evidence bar	permissive · majority · strict	3	forced by this data, §4.4
Missing data	excluded · worst case · capped	3	Bloomberg pp.23, 26
Aggregation	arithmetic · power $p=1/2$ · geometric	3	MSCI p.10 vs Bloomberg p.25

$3 \times 3 \times 2 \times 3 \times 4 \times 3 \times 3 \times 3 \times 3 = 17,496$ specifications, evaluated at any pillar weighting the user selects.

The **elasticity** normaliser deserves note because it is Bloomberg's actual model rather than a variant we invented. They fit, per peer group,

$$\begin{aligned} \log(\text{Impact}) &= \log(\text{Intensity}) + \gamma \cdot \log(\text{Activity}) + \varepsilon \\ \text{Score} &= 10 \times [1 - \Phi(\varepsilon / \sigma)] \end{aligned}$$

and score the residual — how far a company sits from the emissions its peer group's size-elasticity predicts. Their own text notes that when $\gamma = 1$, *the traditional assumption*, ranking on the residual is equivalent to ranking on intensity (Bloomberg, p.46). The entire difference between the market's headline model and a naïve intensity ranking is one parameter that they estimate and everyone else assumes.

4.3 The formula

Nine axes, eight steps. Steps 1–5 produce a grade tensor and do not involve the pillar weights, which is what allows the last steps to run live in the browser. Full statement in **Appendix A**.

1	exposure	divide by the base (quantities) or select the published variant (rates)
2	window	latest reported · fixed fiscal year · 3-year mean
3	direction	level or change
4	orientation	apply polarity; higher = worse, always
5	normalisation	raw value → 0–100 grade within the peer set
6	aggregation	fields → pillar → composite, under the missing-data rule
7	rank	percentile of the composite across the 18
8	distribution	median and interquartile width across the selected specifications

The order of operations matters and must not be inverted. Aggregate the fields *inside* each specification (step 6), *then* take the distribution across specifications (step 8). Reversing them averages 139 fields first, which concentrates the result and collapses the spread — same inputs, opposite conclusion, purely from the order.

Two rules are rules, not specifications, and are not varied: a peer group below **three** members is graded against the sector (two members can only ever return 0 and 100), and the elasticity normaliser requires **eight** points to fit a slope, below which that cell falls back to log min–max.

4.4 The evidence bar — an axis this data forced

The missing-data question has a second form that neither provider documents: not *what do we do with an empty cell*, but *does this field belong in the model at all*. We make it an axis, tightening two things together — how much of the peer set must report a field, and how substantive a policy flag must be.

EXHIBIT 8 The inclusion ladder (fields in the model)

Level	Peer set must report	Boolean tier	E	S	G	F
permissive	≥ 50%	any disclosed policy	28	12	78	9
majority	≥ 67%	constrains an operating decision	12	7	53	8
strict	100%	carries a quantified commitment	2	0	47	8

Sector peer set. Source: *spec.inclusion_mask*.

At the strict end the Social pillar has no fields left, drops out, and the weights renormalise over the three that remain.

Narrowing the peer group runs the other way. For an Exploration & Production company the same strict setting leaves **13 environmental fields and 2 social**, because its peers all report the same things. **Comparing fewer companies lets you compare them on more**. That is the intra- versus inter-group trade, and in our tool it is a control rather than a footnote.

4.5 What we deliberately do not vary

Pillar weights are not an axis. Varying them forms convex combinations of the same indicators; every result lies in the convex hull of the indicator ranks and many draws concentrate on the equal-weighted answer by the central limit theorem. Berg et al. put weights at 13.2% of divergence. Our axes attack the other 86.8%.

Weights are instead handed to the user as a continuous slider with five named presets — a capability, we note, that **neither MSCI nor Bloomberg ships**. Both hand you one vector per industry and no way to change it.

5. Results

5.1 The spread is real and survives its own tail

EXHIBIT 9 Median contestation by stratum

Stratum	Specifications	Median IQR (pts)	Median IQR (places of 18)
Full grid	17,496	41.2	7.0
Level only	8,748	29.4	5.0
Level + majority evidence bar	2,916	35.3	6.0
Level + majority + no worst-case	1,944	32.4	5.5
Level + majority + sector peers	972	23.5	4.0
Level + majority + no worst-case + sector	648	21.3	3.6
Level + strict evidence bar	2,916	17.6	3.0

Remove every branch a critic could call extreme and the median company still moves **3.6 places out of 18**. The contestation is not an artefact of one silly option; it is a property of the evidence.

5.2 What is actually settled

A total ordering of eighteen companies is a claim this data cannot support. What survives is a **partial order**: for each pair, the share of specifications in which one company outranks the other.

EXHIBIT 10 Settled pairwise comparisons

Criterion	Settled	of 153	Share
Full grid, $P > 0.99$	1	153	1%
Full grid, $P > 0.95$	2	153	1%
Full grid, $P > 0.90$	3	153	2%
Full grid, $P > 0.80$	26	153	17%
Full grid, $P > 0.75$	42	153	27%
Level only, $P > 0.95$	33	153	22%
Level + strict, $P > 0.95$	56	153	37%

A **published league table asserts all 153 comparisons**. Under the full grid, two of them hold. Narrow the question to *level* and raise the evidence bar to *strict* and 56 hold — the same data, a narrower question, and a ranking that is partly real emerges from one that was not.

This is the central deliverable. Not a score, but a map of which comparisons the evidence can carry.

5.3 Where the disagreement lives

The grid is a complete balanced factorial, so the variance of a company's rank decomposes into main effects with no design correction: $\eta^2 = SS(\text{axis}) / SS(\text{total})$.

EXHIBIT 11 Variance decomposition, mean across the 18 companies

Axis	Mean η^2	Max for a single company	Companies for which it is the pivot
Direction	18.2%	60.2% (EOG)	9
Peer set	13.1%	31.9% (DVN)	6
Evidence bar	3.8%	12.1%	0
Window	3.8%	20.2%	1
Normaliser	3.3%	12.6%	0
Missing data	3.0%	10.8%	2
Exposure base	1.8%	4.8%	0
Polarity	0.5%	3.4%	0
Aggregation	0.3%	2.2%	0
Main effects	47.8%		
Interaction	52.2%		

Three readings.

The two axes that matter are scope axes. *Direction* is what question you are asking — where a company stands, or how far it has moved. *Peer set* is who you ask it against. Neither is a weight, and neither is a measurement refinement. This is Berg et al.'s 36.7% scope bucket, reproduced from the bottom up on one sector.

More than half the variance is interaction. Main effects account for 47.8%. The remaining 52.2% consists of choices that only matter in combination — which is precisely why sensitivity analysis one dimension at a time, the standard industry practice, understates the problem.

Two axes are negative results and we report them as such. The aggregation function explains 0.3%. This is the fork between MSCI's arithmetic mean and Bloomberg's power mean, a genuine published disagreement with real conceptual content — whether excellent governance can offset poor emissions — and on this data it is worth a third of one percent. The governance-polarity question, to which our own question set devotes an entire worksheet, explains 0.5%. Both were implemented so that we could measure them rather than assume them.

5.4 The remedy — which silence costs the most

For a contested company we can identify the unreported field whose absence carries the most spread, by removing that single silence from the model and measuring how far the interquartile width falls.

EXHIBIT 12 Highest-value missing disclosures

Company	Field	Contestation removed (pts)
TRGP	Total Recordable Incident Rate	29.4
TRGP	Physical Risk Identified	29.4
TRGP	Hazardous Waste Management Policy	29.4
SLB	Percentage of Hazardous Waste	23.5
SLB	Hazardous Waste	23.5
WMB	Total Recordable Incident Rate	17.6
DVN	Tier 1 Process Safety Event Rate	17.6

This converts the framework from a scoring tool into a **targeted disclosure demand**, and names the user the challenge asks us to name: an engagement team, a regulator drafting a disclosure rule, or a company deciding what to publish next.

6. Risks to this analysis

Stated before anyone asks, and in descending order of how much they hurt.

6.1 The grid is enumerated, not sampled. The 17,496 specifications are a set of conventions we constructed, not a random sample from a population of methods. The spread is a *range*, and we never call it a confidence interval. Nor are the cells independent: `change × latest` and `change × mean3` share most of their underlying data, so the effective number of distinct methods is materially below 17,496.

6.2 N = 18 quantises the output. A rank percentile over 18 companies takes 18 values, 5.88 points apart. Any interquartile width quoted in percentile points is a quantised quantity wearing a decimal, which is why every headline in this paper is in rank places. The remedy is a larger universe, not a different statistic.

6.3 The min-max range is not informative at the full grid. Seventeen of eighteen companies span 0 to 100. Those extremes are rare — 2.2% of cells for XOM — and concentrated in `direction = change` crossed with the log-family normalisers and the smallest peer groups. We therefore lead on the interquartile width and report the range only for completeness.

6.4 The economic denominator is a fallback of a fallback. No revenue, no headcount. EBITDA is a defensible scale proxy but it is volatile for this sector and it is not what Bloomberg would use.

6.5 Governance dominates by field count. 79 of 139 fields are governance, and 63 of those are reported by all 18 companies. The G pillar score is consequently far more stable across specifications than E or S, so the contestation we measure is mostly environmental and social. That is a property of the disclosure regime, not of the companies.

6.6 We rank the energy sector against itself. If every constituent is bad, the best relative performer still scores well. Bloomberg names this failure mode in its own methodology (p.42) and does not treat it; we pair every rank with an absolute intensity for the same reason, but relative scoring remains relative.

6.7 Controversies are absent. A whole scoring component present in MSCI and absent from Bloomberg is also absent from us. This is a scope boundary, declared, not an oversight.

6.8 One sector, one year of financials. No cross-sector claim is available from this run, and no trajectory claim is available for the financial pillar.

7. What this is, and what it is not

7.1 It is a framework, and a superset of a rating

The sharpest attack on this work is that it critiques ESG rather than delivering the framework the challenge asked for. It delivers both. Per company the output is a **consensus score** — the median across defensible methods, which is the number the challenge asked for — and a **contestation figure**, which is the part nobody else provides. Anyone who wants one number can take the consensus and ignore the rest. Dropping the second number would not make this more of a framework; it would make it a less honest one.

7.2 It scales

The method is $O(\text{cells})$ and the cells are cheap. The full 17,496-specification run over 18 companies and 139 fields completes in **18 seconds** on a laptop. A secondary pipeline in `ETHack/contested/` demonstrates the data path at sector breadth: 30 companies across 10 GICS sectors, assembled from SEC XBRL frames, EPA GHGRP facility emissions and EPA EEIO supply-chain factors — all free, all government, no API key, roughly 39 API calls for the whole S&P 500 because the bulk endpoints return every filer at once. The binding constraint on scaling is parent-company name matching, not computation.

7.3 Section 8 — net-zero portfolio allocation

The bonus question is answered as an output of this framework rather than as a separate opinion, and is treated in a companion section not included in this draft.

APPENDIX A · THE FORMULA

Indices: company $i = 1 \dots 18$, field $f = 1 \dots 139$, year $y = 1 \dots 5$, base slot $b \in \{1, 2, 3\}$, pillar $P \in \{E, S, G, F\}$.

A.1 Inputs. $\text{VAR}[i, f, y, b]$ the reported number, NaN where not reported; $\text{BASE}[i, b]$ the exposure denominator; $\text{type}[f] \in \{Q \text{ quantity, } R \text{ rate, } P \text{ percentage, } B \text{ boolean, } S \text{ structural}\}$; $\text{pol}[f] \in \{+1, 0, -1\}$ under each of three polarity conventions; $\text{pil}[f]$ the pillar.

A.2 Step 1 — exposure. One operator per type, so the divisor is applied exactly once on every branch:

$x[i, f, y] = \text{VAR}[i, f, y, b] / \text{BASE}[i, b]$	if $\text{type} = Q$	APPLY the divisor
$= \text{VAR}[i, f, y, b]$	otherwise	SELECT the published variant

Slot meanings: for Q, EBITDA / equity / production; for a rate triple, employees / contractors / total workforce; for agency-adjusted leverage, Moody's / S&P / Fitch.

A.3 Steps 2–3 — window and direction. Reductions are taken over the years a company actually reported, not over calendar columns:

level · latest	$q = \text{last reported value}$
level · fiscal year	$q = \text{value at the last complete sector year (FY2024)}$
level · mean3	$q = \text{mean of the last three reported values}$
change · w	$q = (\text{end} - \text{start}) / \text{start} , \text{ end and start per } w$

For booleans, *change* is the plain difference. For single-year fields (the financial pillar) *change* returns NaN and the field drops out.

A.4 Step 4 — orientation. $u[i, f] = \text{pol}[f] \cdot q[i, f]$, so higher u is worse, always. A polarity of 0 means the field is deliberately neutral under that convention: present in the model, discriminating nothing.

A.5 Step 5 — normalisation. Peer set $G(i, p)$; $U = \{u[j, f] : j \in G, \text{ reported}\}$; $m = |U|$. All four produce 0–100 with 100 = best.

percentile	$r = \text{average rank of } u \text{ in } U \text{ ascending};$	$s = 100 (m - r) / (m - 1)$
winsorised	$lo, hi = 5\text{th}, 95\text{th pct of } U;$	$s = 100 (hi - \text{clip}(u, lo, hi)) / (hi - lo)$
log min-max	$L = \log(u \text{ shifted positive});$	$s = 100 (\max L - L) / (\max L - \min L)$
elasticity	$\log u = c + \gamma \log(\text{base}) + \epsilon;$	$s = 100 (1 - \Phi(\epsilon/\sigma))$

A.6 Step 6 — aggregation. Fields $\rightarrow$ pillar $\rightarrow$ composite, both levels using the same weighted shifted power mean with $p = \frac{1}{2}$ and, on this 0–100 scale, $s = 10$:

$$M(x, w, p, s) = (\sum w_j (x_j + s)^p)^{1/p} - s$$

$p = 1$ recovers MSCI's arithmetic mean; $p \rightarrow 0$ the geometric mean. Missing-data rules: **excluded** drops the field and redistributes weight; **worst case** enters it as 0; **capped** drops it and then limits the pillar by its own disclosure factor, $UT = 30 + \sqrt{DF} \times 70$, which is Bloomberg's Issue Score on our scale.

A.7 Step 7 — rank. $R = 100 (N - \rho) / (N - 1)$ where ρ is the average rank of the composite descending. 100 = best of those with a score.

A.8 Step 8 — the two outputs. Over the selected specifications: $\text{consensus} = \text{median } R$, $\text{contestation} = P75(R) - P25(R)$, reported in rank places by dividing by $100/(N - 1) = 5.88$.

APPENDIX B · AXIS OPTION DEFINITIONS

Axis	Option	Definition
Exposure base	ebitda	Q divided by FY2025 EBITDA; R takes the employee-population variant; leverage takes Moody's adjustment
	equity	Q divided by total equity incl. NCI; R takes the contractor variant; leverage takes S&P
	production	Q divided by MMboe/a; R takes total-workforce; leverage takes Fitch. Producers only (9/18)
Window	latest	each company's most recent reported year for that field
	fy_common	FY2024 for every company, the last year the sector reported at full rate
	mean3	mean of the last three reported values
Direction	level	the reduced value itself
	change	relative change from three reported steps earlier; plain difference for booleans
Polarity	orthodox	governance orthodoxy: short tenure, low pay, board refreshment
	neutral	the 15 contested fields score 50 and discriminate nothing
	stability	the continuity reading: retained experience, institutional knowledge
Normaliser	percentile / winsorised / log min-max / elasticity	see A.5
Peer set	subgroup	the five BECS peer groups, with the MIN_PEERS = 3 fallback
	chain	producers (9) vs midstream, refining and services (9)
	sector	all 18
Evidence bar	permissive / majority / strict	see Exhibit 8
Missing data	excluded / worst case / capped	see A.6
Aggregation	arithmetic / power $p=1/2$ / geometric	see A.6

APPENDIX C · FIELD REGISTRY

139 logical fields. Rate triples (TRIR, LTIR, fatality rate) are collapsed to one logical field each so that safety is not counted three times, with the exposure base selecting the published population.

Pillar	Boolean	Percentage	Quantity	Rate	Structural	Total
Environment	18	4	15	0	0	37
Social	9	0	0	5	0	14
Governance	25	0	0	0	54	79
Financial	0	0	0	9	0	9
Total	52	4	15	14	54	139

Boolean stringency tiers: 29 *all only*, 18 *substantive*, 5 *outcome-linked*; the remaining 87 fields are non-boolean and unaffected. Fifteen fields carry a contested polarity — the two conventions disagree about which direction is better — and are concentrated in governance (CEO and chair tenure, board age, pay, auditor tenure, ownership thresholds).

Policy flags versus outcomes. 52 of our 139 fields are yes/no policy flags, or **37%**; the other 87 carry a number. The split is uneven by pillar — environment 18 of 37, social 9 of 14, governance 25 of 79, financial 0 of 9 — so the social pillar in particular leans on policy disclosure more than we would like, and says so.

For comparison, MSCI's own front matter describes 150 policy/programme metrics against **20 performance metrics** on the management side, a 7.5:1 policy-to-outcome ratio (MSCI p.2). Bloomberg pushes the other way and builds machinery to enforce it: the disclosure factor exists to stop a perfect policy score coexisting with zero quantitative disclosure, and an issue containing only binary fields has its weight cut by 80% (Bloomberg pp.8, 24). Our 37% sits between the two, closer to Bloomberg, and the *evidence bar* axis is the control that lets a reader push it further — at the strict setting only the five outcome-linked booleans survive.

APPENDIX D · PROVIDER CITATIONS

- **MSCI ESG Ratings Methodology, Executive Summary, November 2020**, 17pp. Cited as *MSCI p.N*. Key pages: p.2 data volumes; p.3 materiality definition; p.4 pillar/theme/key-issue hierarchy and WAKIS; p.5 key-issue weights and the 33% governance floor; pp.6–8 exposure, management and the risk/opportunity combination; p.9 controversy severity matrix and the absolute governance model; p.10 GICS Sub-Industry weighting; p.12 industry benchmark percentiles, the November 2020 change, and the seven-band letter map; p.13 data sources.
- **Bloomberg ESG Scores Methodology, issued September 2023, updated December 2025**, 58pp. Cited as *Bloomberg p.N*. Key pages: p.7 BECS; p.8 financial materiality and score structure; p.11 input data and imputation; p.17 field attributes, disclosure-factor ranks and activity metrics; p.18 ES scoring models; p.21 issue priority; p.23 fit/quality weights; pp.24–26 the disclosure factor and the Issue Score cap; p.28 the priority weight function; p.32 pillar weights and worked industry examples; p.34 Latest vs Fiscal Year; p.41 parametric versus percentile; p.42 absolute versus relative and the best-in-class failure mode; pp.45–46 the intensity model and the $\gamma = 1$ remark; pp.47–49 the remaining field-scoring distributions.
- Berg, F., Kölbel, J. F. & Rigobon, R. (2022). *Aggregate Confusion: The Divergence of ESG Ratings*. *Review of Finance* 26(6).
- Simonsohn, U., Simmons, J. P. & Nelson, L. D. (2020). *Specification curve analysis*. *Nature Human Behaviour* 4, 1208–1214.
- Dimson, E., Marsh, P. & Staunton, M. (2020). *Divergent ESG Ratings*. *Journal of Portfolio Management* 47(1).

APPENDIX E · REPRODUCIBILITY

```
cd ETHack/model
python3 registry.py      # the field registry, printed for inspection
python3 load.py          # the arrays, with a coverage report
python3 run.py           # all 17,496 specifications -> out/
python3 export_web.py    # the pillar cube -> ../web/data/
python3 -m http.server 4173 --directory ../web
```

numpy and pandas only. Runtime 18s for the full grid.

Output	Contents
out/summary.csv	consensus, contestation, verdict, coverage per company
out/long.csv	one row per company per specification — the audit trail
out/pivot.csv	variance decomposition per company, per axis
out/dominance.csv	$P(i \text{ outranks } j)$ for all 153 pairs
out/remedy.csv	highest-value missing disclosures
out/claims.json	every number quoted in this paper, computed

The browser explorer reproduces `run.py` to 0.05 rank points on all 18 companies, and the specification-index decode is verified against Python cell by cell, so nothing in the interactive tool is precomputed toward a particular answer.

Data withheld. Bloomberg Terminal cell values are not redistributed; see `bruh/mat maps/WITHHELD.md` for the manifest and the regeneration path. We do not share these indicators and values because they require the proper licensing. We are happy to show the files in person, to demonstrate that the model is built on real data and that the approach is scalable and feasible. Every derived output — the loader, the formula, all computed results, the explorer — is published.

APPENDIX F · GLOSSARY

Term	Meaning
Consensus	Median rank percentile across the specifications in view
Contestation	Interquartile width of that distribution, in rank places out of 18
Specification	One complete set of choices across all nine axes
Pivot	The axis explaining the largest share of a company's rank variance
Settled pair	An ordered pair holding in >95% of specifications both companies score in
Evidence bar	The joint threshold on peer-set reporting share and boolean substantiveness
Disclosure factor	Share of expected fields a company reports; caps the pillar under the <i>capped</i> rule
BECS	Bloomberg ESG Industry Classification System, their purpose-built ESG peer grouping
WAKIS	Weighted Average Key Issue Score, MSCI's pre-normalisation composite

Disclosures

Prepared for ETHack 2026, Challenge #1, sponsored by Citadel and Citadel Securities, by a student team over 24 hours. **This is not investment research, it is not a recommendation to buy or sell any security, and it was not prepared by a regulated entity.** Company names appear as data points in a methodological demonstration and no view on any issuer is expressed or implied.

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